



Module:	MAT602 + MTRM011	Title:	Materials Selection in Design
First examiner:	Prof James Busfield	Second examiner:	

2013

Thrust deflectors and reversers on aircraft engines are controlled by hydraulic actuators. The forces required to reverse thrust can be large, so the actuators (of which each engine may have four) are heavy. One of the heavier parts is the piston rod, identified in Figure 2. It carries axial compressive loads only, and is designed as a hollow tube, to save weight. The cross section is chosen to carry the loads without compressive failure or buckling.

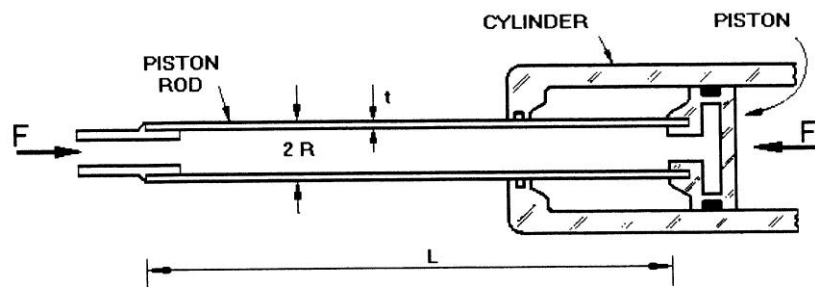


Figure 2. A piston rod

The main objective is to minimise the weight. The length L and the outer radius R are specified. The piston rod must transmit a maximum axial force F , it must not fail by yielding in compression or by elastic buckling and it should operate at a temperature up to 300°C . The piston rod as a thin-walled tube of wall thickness t .

(a) Derive the equation for the mass of the rod.

$$\text{Mass} = 2\pi RLt\rho$$

1 mark

(b) Specify the failure criterion for a yield failure?

$$F/2\pi Rt > \sigma_y$$

1 mark

(c) By eliminating the wall thickness from your two equations, calculate the mass of the piston rod and derive the material performance index M_1 for this minimum mass constraint.

$$\text{Mass} = \rho LF / \sigma_y$$

$$M_1 = \text{low } (\rho / \sigma_y)$$

1 mark

(d) Buckling occurs when an axially loaded strut is compressed by force $F = \pi^2 EI / L^2$ where I is given by $I = \pi R^3 t$ for a cylinder. Calculate an alternative material performance index M_2 for a minimum mass strut that will not buckle under load.

$$t = FL^2 / (\pi^3 ER^3)$$

$$\text{Mass} = 2L^3 F / (\pi R)^2 (\rho / E)$$

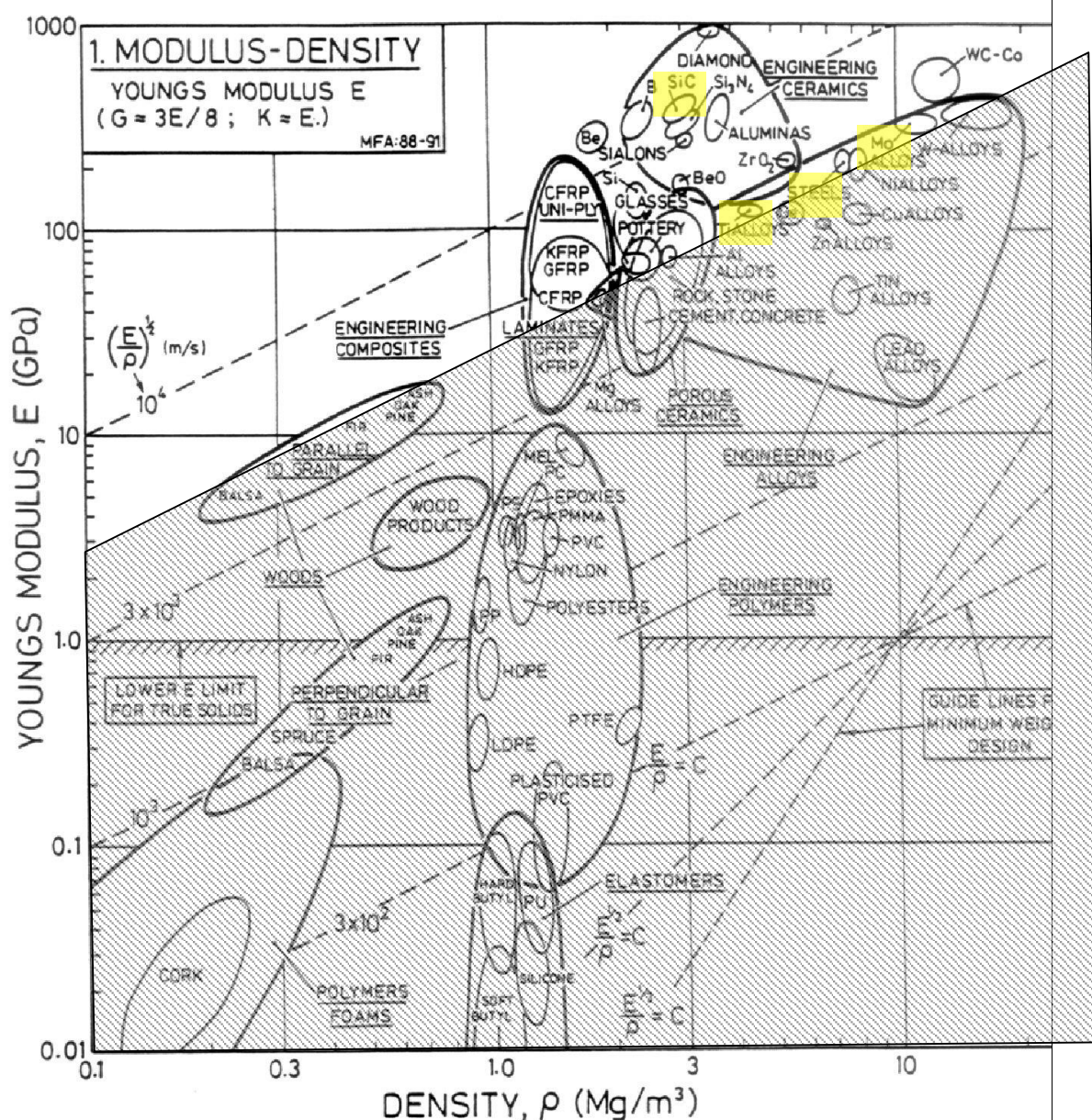
$$M_2 = \text{low } (\rho / E)$$

2 marks

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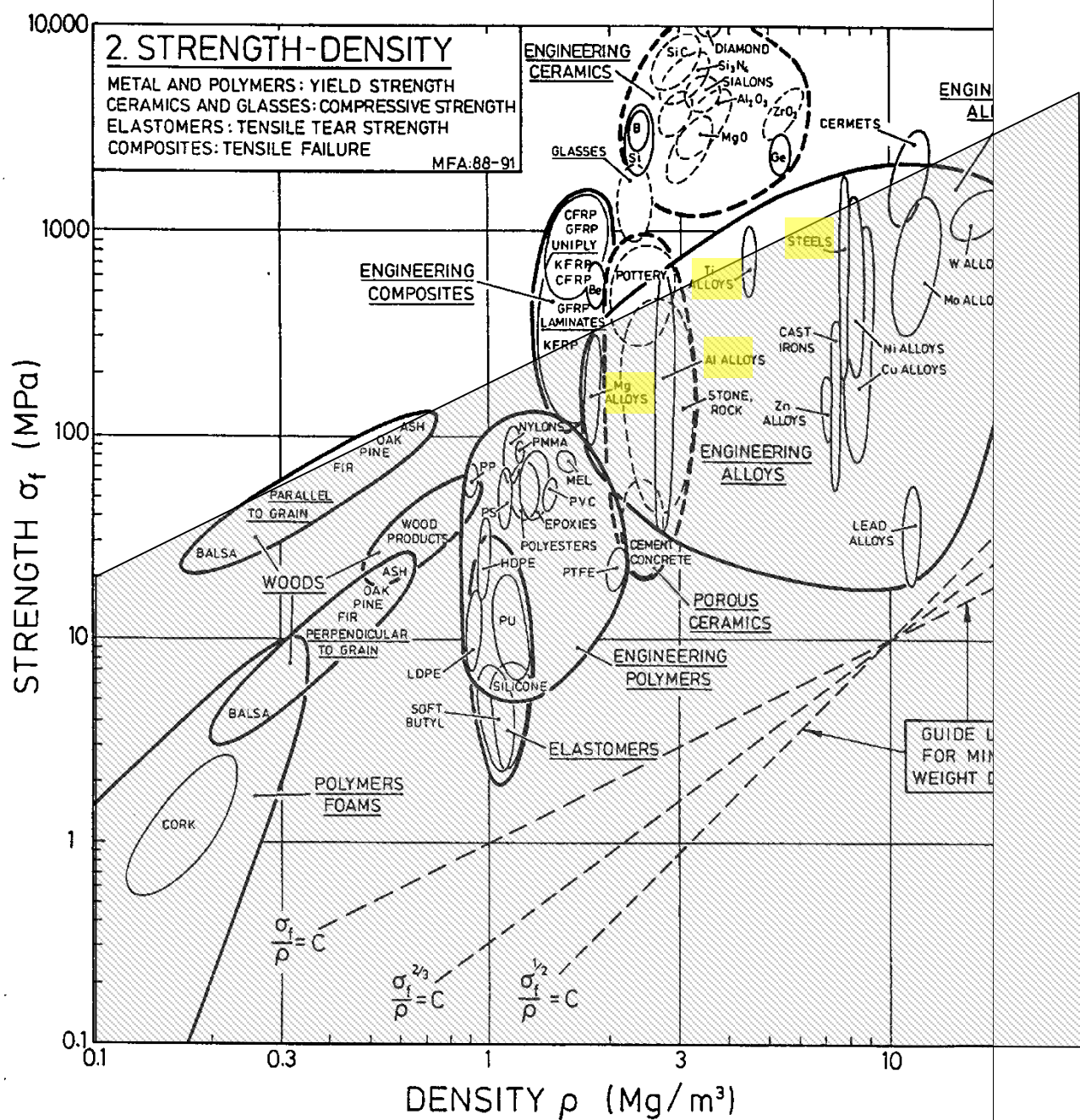
Draw lines on either or both of the graphs that follow to plot both derived performance indices M1 and M2 and use the circles to highlight the most suitable materials for use as a thrust deflector piston rod

5 marks



5 marks

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	Tabulate below three of the most suitable materials for use as a thrust deflectors from the Ashby diagrams in Question 2. Calculate both materials' indices for these materials and describe the suitability of each when selected for use as a thrust deflector.			5 marks
	Selected Material	M1 MPa.m ³ /Mg	M2 GPa.m ³ /Mg	
	Steel	50	25	
	Aluminium	50	25	
	Magnesium	100	30	
	Beryllium alloys	250	60	
	Titanium	200	25	
	Carbon Fibre composites	500	100	
	Silicon Carbide	2000	200	
	Diamond	3000	300	
	SiC reinforced Ti matrix composites are the best.	400	100	



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2014

An oar can be considered to be a strong, light, stiff beam that is loaded in bending. Its basic requirement is that it must carry the bending moment exerted by the oarsman without breaking. Its stiffness is measured by hanging a weight of 10kg on the oar, 2.05m from the collar and the deflection at the point of loading is measured, as shown below.

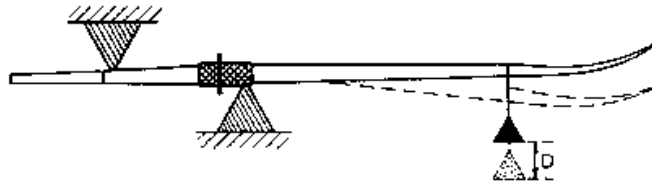


Figure 1.

Assuming that the oar is of uniform cross sectional area, a , has a length L , and a second moment of area, I . The density of the material it is made from is ρ . What is the mass, m of the oar?

Mass, $m = aL\rho$(1)

Assuming that the deflection of a simply supported beam, D when it is subjected to a load F is

$$D = \frac{FL^3}{C_1EI}$$

calculate the stiffness of the beam. S .

$$S = \frac{F}{D} = \frac{C_1EI}{L^3} \dots\dots\dots(1)$$

For a circular cross section $I = \frac{a^4}{32}$. By eliminating the cross section area, a derive an equation for the mass of the oar.

$$S = \frac{C_1Ea^4}{32L^3}, a = \sqrt[4]{\frac{32SL^3}{C_1E}} \dots\dots\dots$$

$$m = \left(\frac{32S}{C_1}\right)^{1/4} (L)^{3/4} \left(\frac{\rho}{E^{1/2}}\right) \dots\dots\dots(1)$$

Separate this equation up into functional, geometric and materials components.

$$m = \left(\frac{32S}{C_1}\right)^{1/4} (L)^{3/4} \left(\frac{\rho}{E^{1/2}}\right) \dots\dots\dots(1)$$

Identify what the materials index is that optimises the design of the oar

$$M_1(\text{minimum}) = \left(\frac{\rho}{E^{1/2}}\right) \dots\dots\dots(1)$$

5

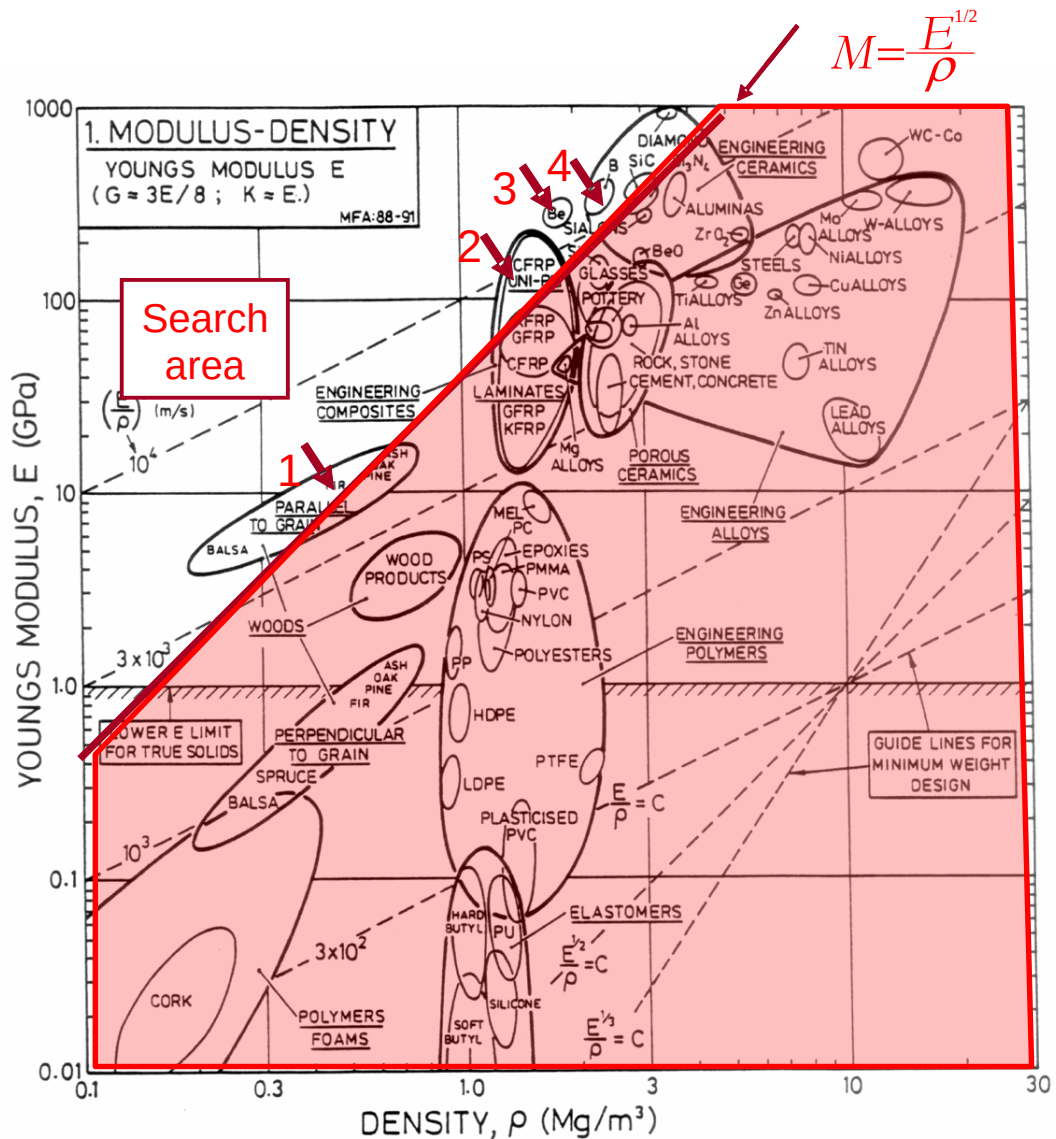


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2

Using the Ashby diagram below, highlight the most suitable materials from a range of materials classes that can be used as a minimum mass for the oar described in question 1.

3



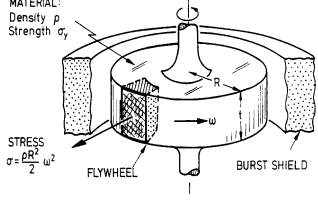


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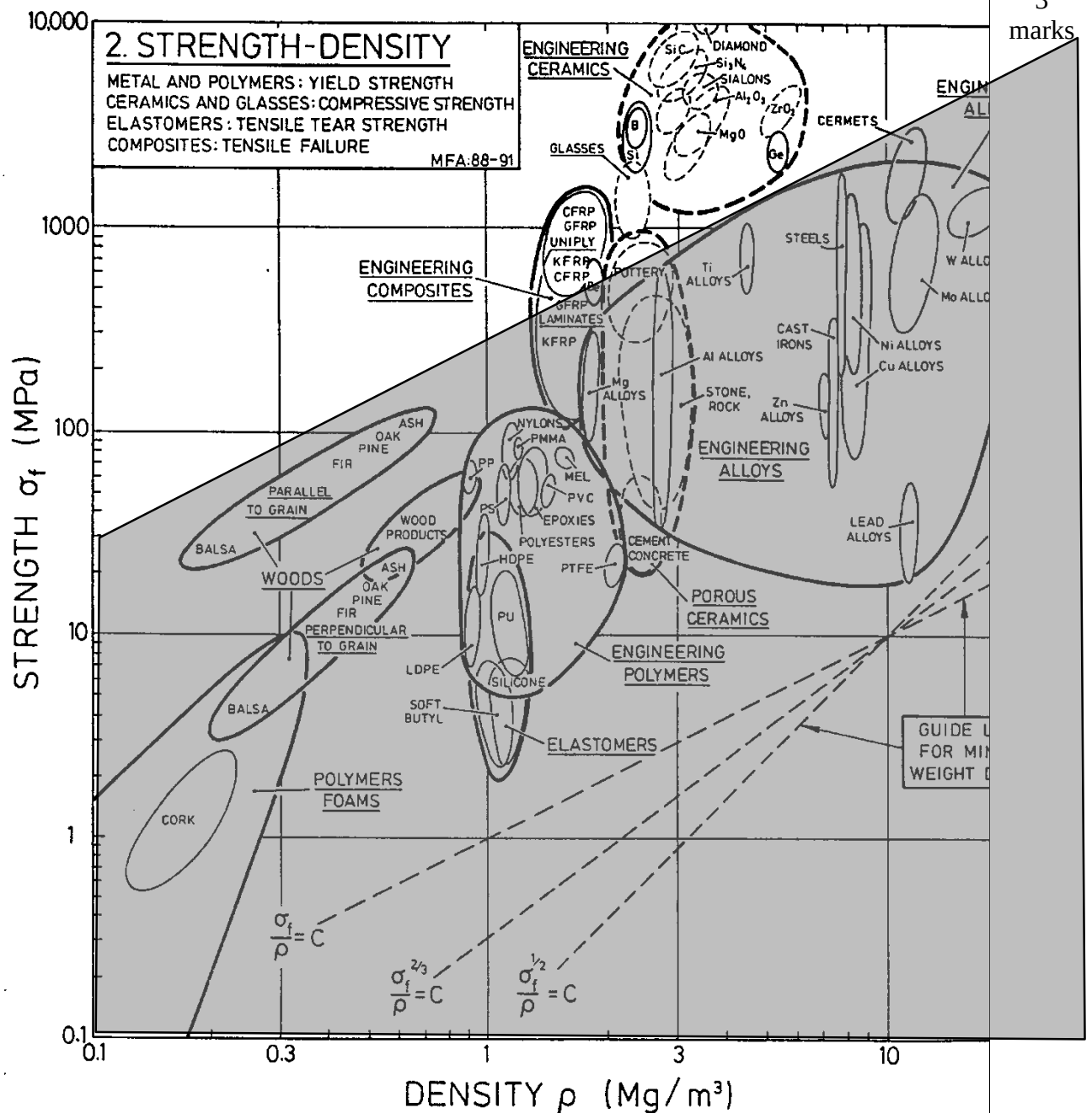
3	Tabulate below the top five materials for use as an oar from the Ashby diagram in Question 2. Calculate the materials' index for that material and describe the suitability of the materials selected for use as an oar.	4																								
	<table><tr><th>Selected Material</th><th>Index / Units</th><th>Could also be $\text{GPa}^{0.5}/\text{Mgm}^3$</th><th>Comments on suitability</th></tr><tr><td>Wood</td><td>$150\text{kg}/(\text{m}^3\text{GPa}^{0.5})$</td><td>7</td><td>Cheap, light, but variable</td></tr><tr><td>CFRP</td><td>$100\text{kg}/(\text{m}^3\text{GPa}^{0.5})$</td><td>10</td><td>The best choice, more control of design.</td></tr><tr><td>Beryllium</td><td>$120\text{kg}/(\text{m}^3\text{GPa}^{0.5})$</td><td>8</td><td>Cost (and toxicity) rule it out</td></tr><tr><td>Ceramics</td><td>$200\text{kg}/(\text{m}^3\text{GPa}^{0.5})$</td><td>5</td><td>But fracture toughness inadequate</td></tr><tr><td>Diamond</td><td>$100\text{kg}/(\text{m}^3\text{GPa}^{0.5})$</td><td>10</td><td>Ridiculous choice</td></tr></table>	Selected Material	Index / Units	Could also be $\text{GPa}^{0.5}/\text{Mgm}^3$	Comments on suitability	Wood	$150\text{kg}/(\text{m}^3\text{GPa}^{0.5})$	7	Cheap, light, but variable	CFRP	$100\text{kg}/(\text{m}^3\text{GPa}^{0.5})$	10	The best choice, more control of design.	Beryllium	$120\text{kg}/(\text{m}^3\text{GPa}^{0.5})$	8	Cost (and toxicity) rule it out	Ceramics	$200\text{kg}/(\text{m}^3\text{GPa}^{0.5})$	5	But fracture toughness inadequate	Diamond	$100\text{kg}/(\text{m}^3\text{GPa}^{0.5})$	10	Ridiculous choice	
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2015	Part C Question 1. Flywheels store energy, small ones used in toy cars are made of lead. During the industrial age steam engines had flywheels made from cast iron. Most recently flywheels have been proposed for energy storage. What is the best material to choose for the wheel?  The efficient flywheel stores as much energy per unit weight as possible, without failing. It can be considered as a solid of radius R and thickness t, with a material of density ρ rotating with an angular velocity of ω. What is the equation for the kinetic energy stored in a flywheel, U, expressed in terms of the moment of inertia, I? $U = \frac{1}{2} I \omega^2$ _____ (1) Given that the moment of inertia for a solid cylinder about its central axis is given as, $I = \frac{1}{2} \pi R^4 t \rho$, derive the equation for the kinetic energy in terms of the flywheel geometry and density. $U = \frac{1}{4} \pi R^4 t \rho \omega^2$ _____ (1) Calculate the mass of the flywheel, m, expressed in terms of the geometry and the material density. $m = \pi R^2 t \rho$ _____ (1) The flywheel must not burst and so the hoop stresses, σ generated by the rotation must not exceed the yield strength of the material, σ_y $\sigma = \frac{(3 + \nu)}{8} \rho R^2 \omega^2 \leq \sigma_y$ By eliminating the free variables, identify the materials parameter that can most readily be used to identify the best material to make a light weight energy storing flywheel. $U/m = \frac{1}{4} \pi R^4 t \rho \omega^2 / \pi R^2 t \rho = \frac{1}{4} R^2 \omega^2$ $R^2 \omega^2 \leq \frac{8 \sigma_y}{(3 + \nu) \rho}$ $U/m = 2 \sigma_y / \rho (3 + \nu) \approx \sigma_y / 2 \rho$ _____ (2)	5
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What other issues should you consider when selecting a suitable material for making a flywheel?

The energy stored here is very large and the rotor must be shielded with a very strong burst shell to protect the mechanism in the event of a catastrophic failure.

2
marks



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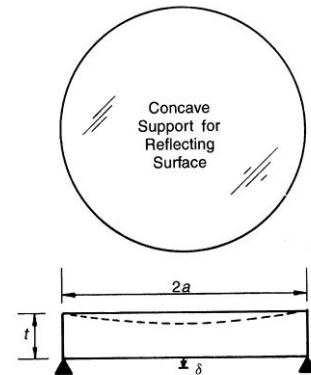
3	3. Tabulate below the top five materials for use as an flywheel from the Ashby diagram given in Question 2. Calculate the materials' index for that material and describe the suitability of the materials selected for use as a flywheel. <table border="1"> <thead> <tr> <th>Selected Material</th><th>Index (kJ/kg)</th><th>Comments on suitability</th></tr> </thead> <tbody> <tr> <td>Ceramics</td><td>200-2000</td><td>It is the wrong strength as they are all too weak in tension</td></tr> <tr> <td>Composites – CFRP</td><td>100-500</td><td>The best choice – but very expensive</td></tr> <tr> <td>Composites – GFRP</td><td>100-400</td><td>Cheaper than CFRP and still very good. The normal choice for electrical storage flywheels.</td></tr> <tr> <td>HS steels</td><td>100-200</td><td>Not bad and significantly cheaper than composites</td></tr> <tr> <td>Ti Alloys</td><td>100-200</td><td>Not bad but much more expensive than steel for little performance gain.</td></tr> </tbody> </table> Other possible answers include <table> <tr> <td>Beryllium</td> <td>250-300</td> <td>Expensive, but very good.</td> </tr> <tr> <td>HS Al alloys</td> <td>100-200</td> <td>Occasionally used in practice due to lowest cost.</td> </tr> <tr> <td>HS Mg alloys</td> <td>100-200</td> <td>Not bad but quite expensive</td> </tr> </table>	Selected Material	Index (kJ/kg)	Comments on suitability	Ceramics	200-2000	It is the wrong strength as they are all too weak in tension	Composites – CFRP	100-500	The best choice – but very expensive	Composites – GFRP	100-400	Cheaper than CFRP and still very good. The normal choice for electrical storage flywheels.	HS steels	100-200	Not bad and significantly cheaper than composites	Ti Alloys	100-200	Not bad but much more expensive than steel for little performance gain.	Beryllium	250-300	Expensive, but very good.	HS Al alloys	100-200	Occasionally used in practice due to lowest cost.	HS Mg alloys	100-200	Not bad but quite expensive	5
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2016

There are some very large optical telescopes in the world. The simplest types adopt a reflector that is a simple concave mirror. The largest of this type have mirrors up to 6m in diameter. The mirrors must be light enough for ease of transportation to the top of mountains. They must also be stiff enough not to significantly deflect more than the wavelength of light under their own weight. The mirror itself is made up of two parts: a reflective front surface and a concave support. This question is concerned identifying the best materials for the concave support.



By considering the geometry and density, ρ , of the material derive an equation for the mass, m , of the cylindrical support for the mirror surface.

$$m = \pi a^2 t \rho$$

If the mirror is supported around the perimeter then the greatest elastic deflection δ at the centre of the mirror is given as $\delta = \frac{3}{4\pi} \frac{mga^2}{Et^3}$, where E is the elastic modulus of the material and g is the acceleration due to gravity. Rearrange this equation to solve for t .

$$t = \left(\frac{3}{4\pi} \right)^{1/3} \left(\frac{mga^2}{E\delta} \right)^{1/3}$$

Derive a simple expression for the mass, m , of the telescope when the free variable is the thickness, t , and the fixed variables are the radius, a .

$$m = \left(\frac{3g}{4\delta} \right)^{1/2} (\pi^{1/4} a)^4 \left(\frac{\rho}{E^{1/3}} \right)^{3/2}$$

Separate out this relationship for the mass into the functional, geometric and materials properties

$$p = f_1(F) f_2(G) f_3(M)$$

Identify the materials index M that must be maximised to make the lightest, stiff mirror support.

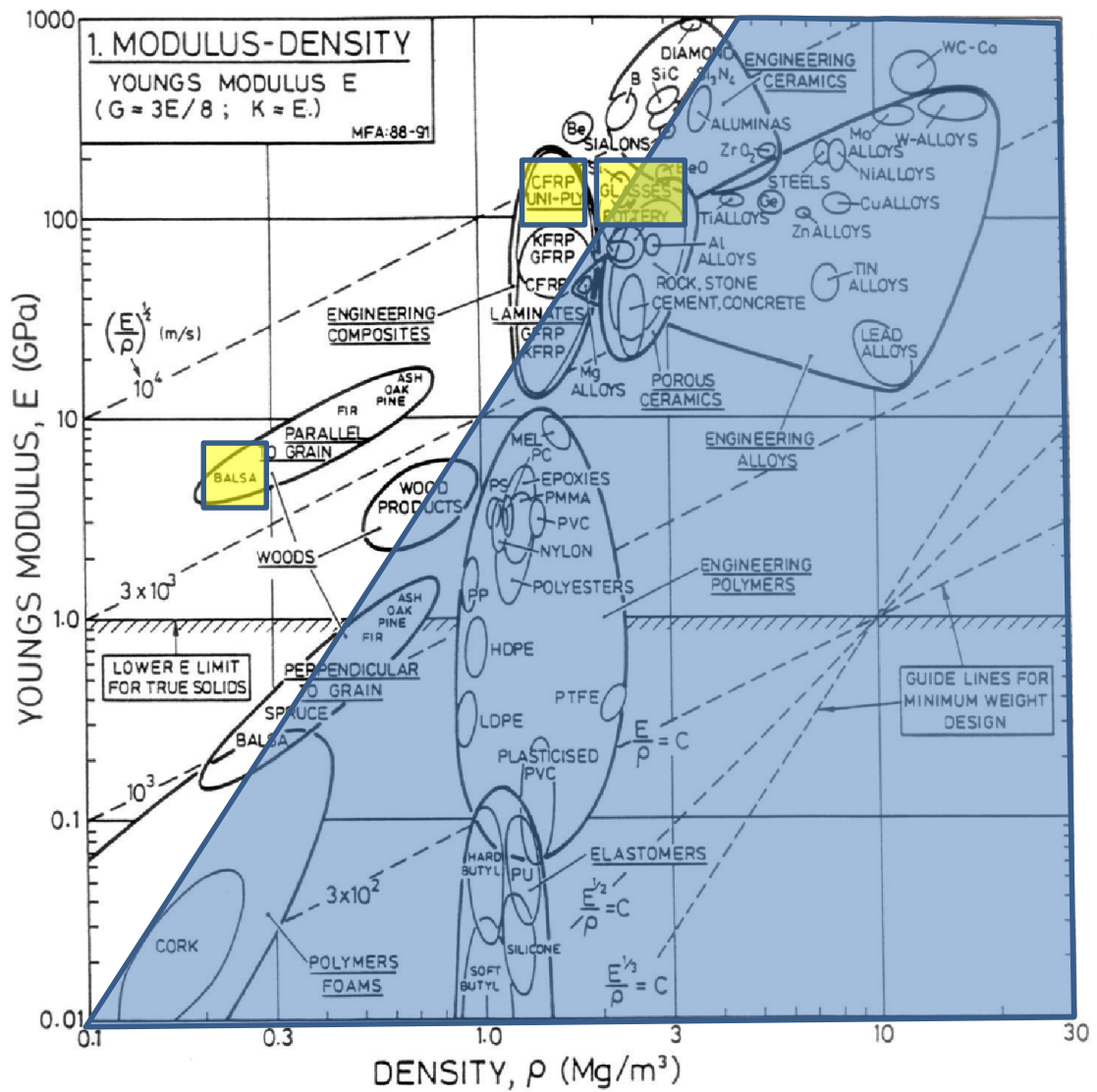
$$M_1 = \left(\frac{E^{1/3}}{\rho} \right)$$

Question
C2

Using the Ashby diagram below highlight the most suitable materials from a range of materials classes that can be used as a concave mirror support identified in Question C1.



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3 for
line,
shading
and
picking
best
options

What other issues should you consider when selecting a suitable material for making a concave mirror support?

2 marks

Will it change shape with temperature, humidity or other with time



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C3

Tabulate below the top 3 materials for use as concave mirror support from the Ashby diagram given in Question **C2**. Calculate the materials' index for that material and describe the suitability of the materials selected for use as a flywheel.

Selected Material	Index / $\text{Pa}^{1/3} \cdot \text{m}^3/\text{kg}$ Identify the Units	Comments on suitability
Glass	2.5	The current material of choice (Extra credit for identifying the foamed glass not on the chart but available in the tutorial using the CES database.
Any of a range of woods (oak / balsa/ pine)	~10	Cannot use as they are not geometrically stable enough with humidity
Composites (CFRP, GFRP.....)	~3	Issues in shape change with temperature and humidity

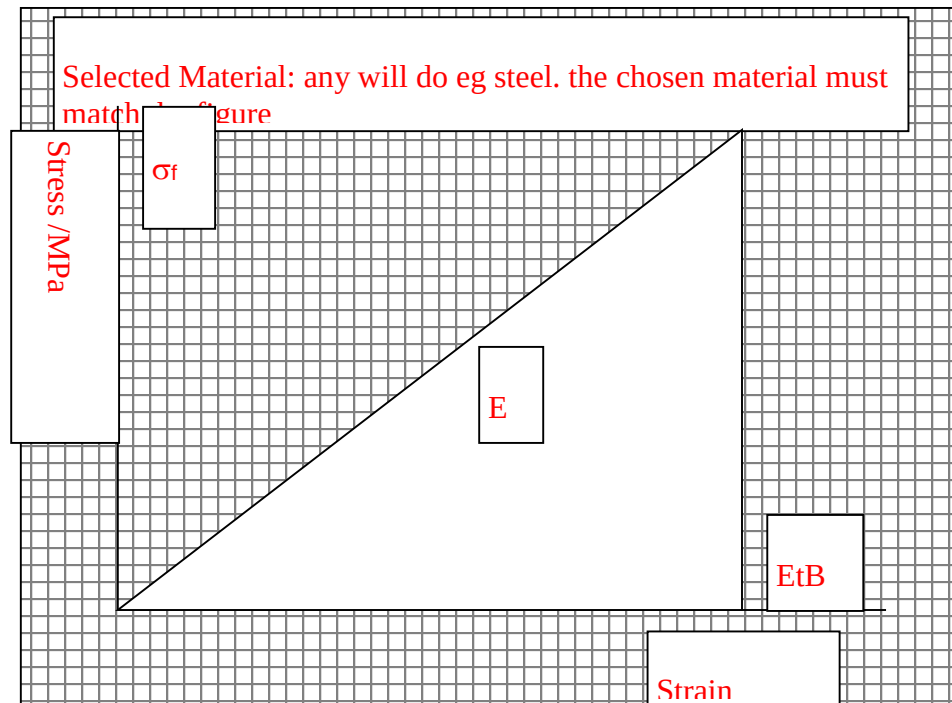
3 marks
in total
or 1 per
material



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2017

Question 1. A spring is a device that stores elastic energy. Sketch a suitable linear elastic stress versus strain graph for any specified material and identify the maximum elastic energy density, W that can be stored in the spring.



2

Derive below the following equation from your figure for the elastically stored energy expressed only in terms of the modulus, E , and the failure strength, σ_f , of the material.

$$W = 0.5 \sigma_f^2 / E$$

Energy stored = W = Area of a triangle = $0.5 \text{ base} \times \text{height} = 0.5 \text{ EtB} \times \sigma_f$

$$E = \sigma_f / \text{EtB}$$

Therefore $W = \frac{1}{2} \frac{\sigma_f^2}{E}$. (2 marks)

2

Using the above relationship identify a suitable materials selection parameter for a spring that has a minimum volume design which can store the maximum energy.

M1 minimum σ_f^2 / E (1 mark)

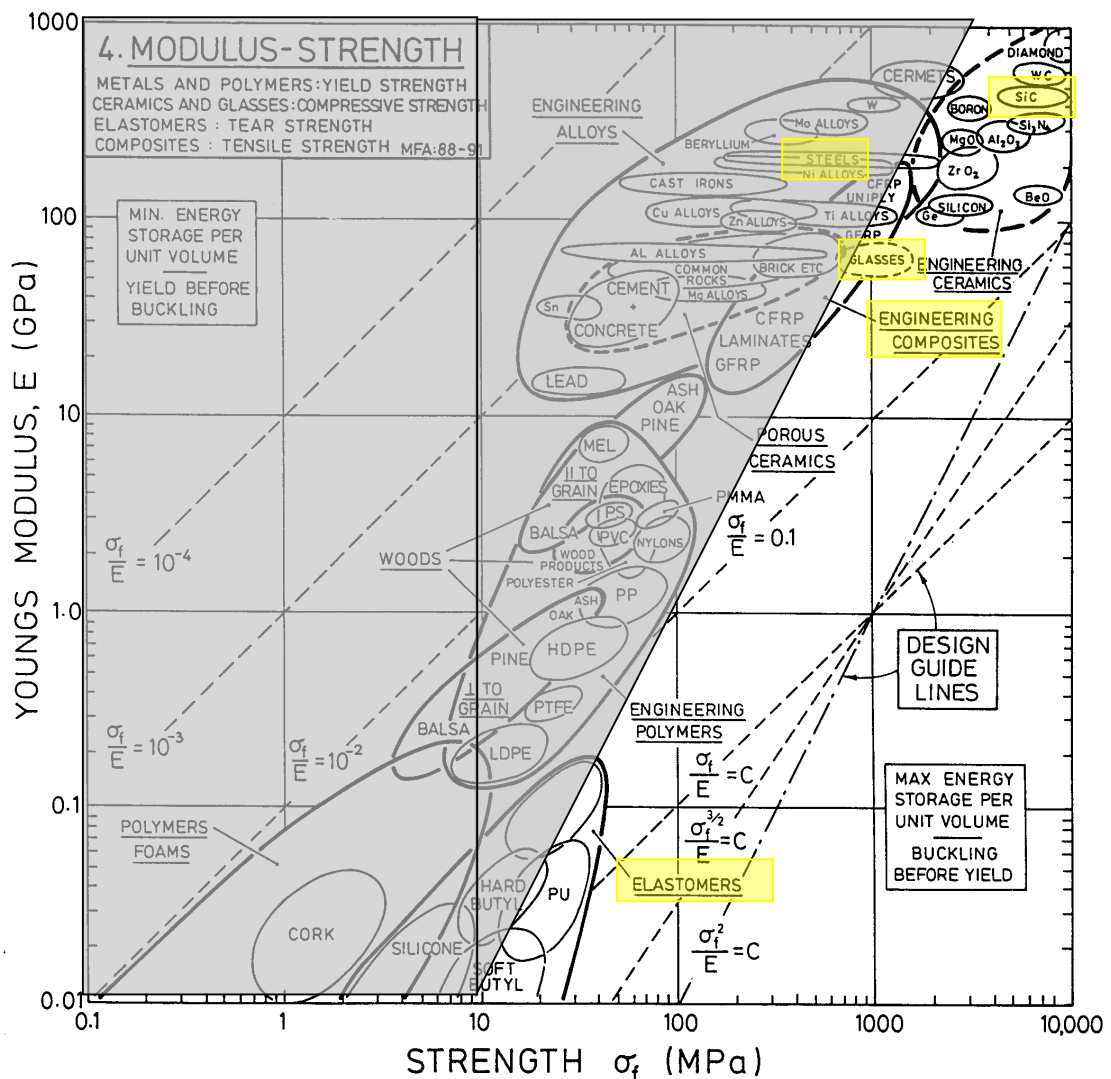
1



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Question
C2

Question 2. Using the Ashby diagram given, identify the most suitable materials from a range of materials classes that can be used as a minimum volume spring material. You should indicate on the chart using lines to define regions of interest, what you might think are the optimum materials and you should highlight your chosen materials.



2 for
line,
shading
and
picking
best
options

2 marks



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Question C3.	Selected Material	Index/ MPa	Comments on suitability	4 marks in total or 1 per material
	Rubbers	20	Good materials but includes some damping, used often in vibration control.	
	Glasses	10	Weak in tension and bending, with very low losses and so used in galvanometers.	
	Carbon Fibre RP	8	Comparable to steel but more expensive.	
	Nylon	3	Low cost and easy to make, used in toys for example.	
	BeO & other eng ceramics	10-100	All weak in tension	
	Glass Fibre RP	5	Similar to CFRP but much cheaper.	
	Spring Steel	10	Very widely used from suspensions in building, trains and planes to watches.	
	Titanium	10	Excellent prestige use material, with good corrosion properties.	



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2018

A flexible elastic hinge, such as that shown in Figure 1, can transmit a load between components whilst allowing a specific motion between them by deforming elastically.

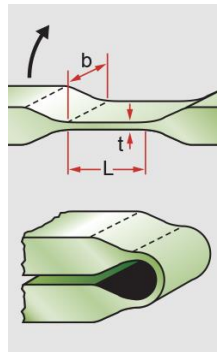


Figure 1 – An elastic hinge

The aim is to maximise the movement (or the angle over which it can be rotated) without resulting in any permanent damage. When considering a box such as a DVD case, then a hinge is formed along one side of the opening. The best material is the one (for a given ligament dimension) that can be bent to the smallest radius without breaking. If a ligament of thickness t is bent elastically to a radius R , then the surface strain is given as

$$\epsilon = \frac{t}{2R}$$

If the hinge is still behaving elastically what is the maximum stress in the hinge?

$E = \sigma / \epsilon$ therefore $\sigma = tE / 2R$

1

The tightest radius that can be achieved is when the materials reaches its yield strength. Therefore, derive the smallest radius that can be achieved for a given material.

$R = tE / 2\sigma_f$

2

Identify the materials index that produces the smallest radius

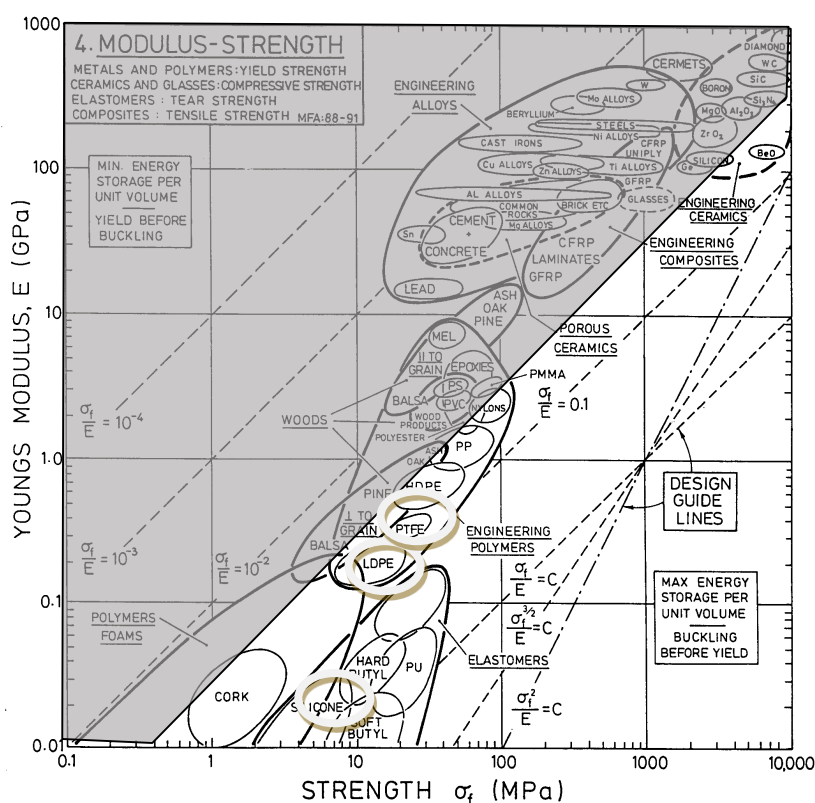
R small therefore E / σ_f is also small

2

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Using the Ashby diagram given below, identify the most suitable materials from a range of materials classes that can be used for an elastic hinge using the relationship derived in Question C1.

2



Marks are awarded for drawing a correctly labelled line with the correct slope, and at the correct position so as to select just a handful of materials. You must shade some areas of the diagram to indicate which materials can be eliminated.

Tabulate below three suitable materials for use as elastic hinge from the Ashby diagram given in Question C2. Calculate the materials' index for that material and describe the suitability of the materials selected for use as a spring.

3 marks

Selected Material	Index / No units Identify the Units	Comments on suitability
Polyethylene	~30	Cheap and widely used for hinged goods and bottle caps etc.
Polypropylene	~30	Very tough and easy to mould. Used widely for DVD cases etc.
Rubbers / Elastomers	~3	Outstanding hinges but often too soft



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Part C
Q1

A reusable elastic seal shown in Figure 1 consists of a cylinder, with a radius R , and Young's modulus, E , compressed between two flat surfaces. The seal must form the largest possible contact width, b , while keeping the contact stress, σ , less than 100MPa so that it does not damage the flat surfaces. The seal itself must remain elastic so that it can be reused many times.

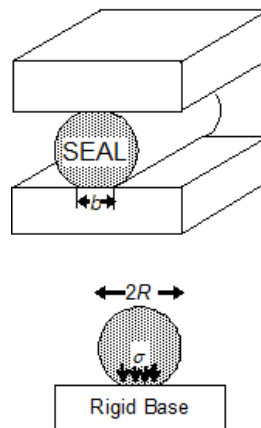


Figure 1.

Assuming that a force f per unit length forms an elastic contact of width, b given by

$$b \approx 2 \left[\frac{fR}{E} \right]^{\frac{1}{2}}$$

and that the contact stress can be approximated by,

$$\sigma = 0.6 \left[\frac{fE}{R} \right]^{\frac{1}{2}}$$

Derive below a suitable material selection index.

Eliminate f and stipulate that $\sigma < \sigma_f$ allows one to derive

$$b \leq 3.3R \left[\frac{\sigma_f}{E} \right]$$

Max b requires max $[\sigma_f/E]$

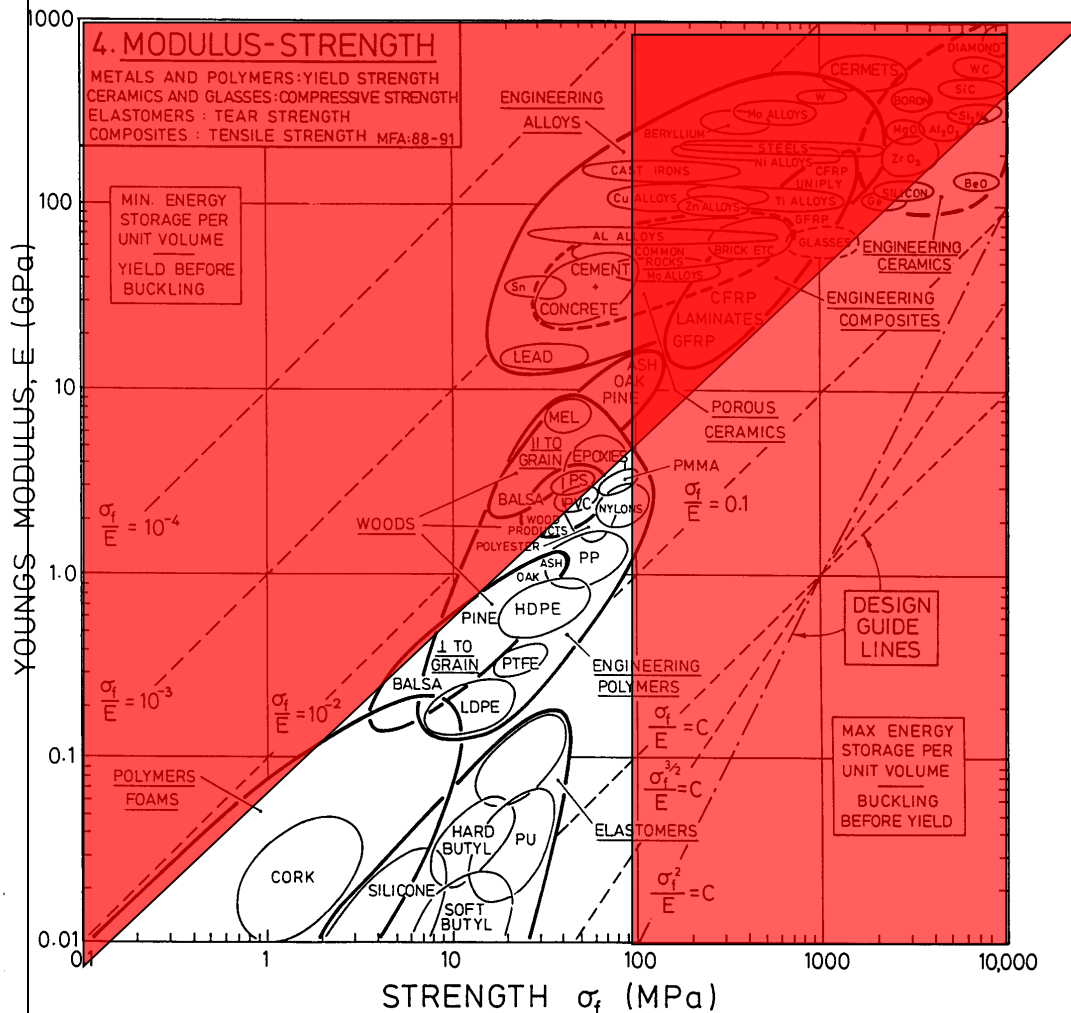
It is important to realise that the stress in the equation has to be turned in to a materials property like the failure strength.

A second criteria is that the contact stress must be less than 100MPa.



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Q2 Plot the relationship found in Question 1 in the diagram below. In addition, the contact pressure must be maintained below 100MPa to prevent plastic deformation of the flat surfaces. Plot this relationship also on the diagram.



Marks are awarded for drawing two correctly labelled lines with the correct slope, and at the correct position so as to select just a handful of materials. You must shade some areas of the diagram to indicate which materials can be eliminated.

(5)



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Q3

Identify the top three materials for use as a seal as suggested from the Ashby diagram in question 2. Calculate the materials' index for that material and discuss their relative merits as sealing materials.

Selected Materials (should be a couple from this list + 1 other)	Index Strength / Modulus - Make both into MPa – gives no units	Comments on suitability
Butyl Rubber	2	Commonly used (poor chemical and heat resistance)
Silicone Rubber	0.5	Similar to butyl but with better temperature and chemical resistance. Not very strong
Polyethylene	0.1	Cheap but not very elastic
Cork	0.1	A traditional choice for seals for centuries, reasonable chemical resistance also. Very low contact stresses also.
Polyurethanes	3	Widely used family of materials some of which have good chemical stable